Nerrylle Faith Tumpalan Pabalan

Calamba, Laguna • pabalan.361525@calamba.sti.edu.ph

EDUCATION

STI College Calamba *Calamba, Laguna*

Bachelor of Science in Computer Engineering *Expected 2027*

Thesis: "Smart Bi-Directional Philippine Currency Exchanger for Bills and Coins" (In Progress)

Relevant Coursework: Microprocessors, Embedded Systems, Logic Circuits and Design, Data Structures and Algorithms, Computer Network and Security, Object-Oriented Programming, Introduction to HDL, Systems Administration, Digital Signal Processing, Network Technology, CpE Laws and Professional Practice

TECHNICAL PROJECTS

Smart Bi-Directional Philippine Currency Exchanger *Calamba, Laguna*

Hardware-Software Lead (Capstone Design 1 & 2) *June 2026 – Present*

- Engineered a full-stack hardware/software kiosk automating bill-to-coin and coin-to-bill exchange, integrating a Raspberry Pi 4 (Flask, OpenCV/ORB currency recognition) with an Arduino Mega for sensor and actuator control.
- Designed a custom 4-module mechanical bill dispenser in Siemens Solid Edge — including gear trains, a roller-based singulation mechanism, and a separation system — validated through iterative prototyping and testing.
- Built a serial communication protocol between embedded and application layers to support real-time transaction status, inventory tracking, and SMS receipt delivery.
- Collaborating with a 3-member team to manage project deliverables, system testing, debugging, and cross-layer hardware-software troubleshooting.

Computer-Aided Engineering Drafting and Design Prototyping *Calamba, Laguna*

Project Developer (Academic Coursework) *Academic Year 2024 – 2025*

- Utilized Siemens Solid Edge to construct accurate parametric 3D models, complex mechanical part assemblies, and production-ready engineering drawings.
- Applied Geometric Dimensioning and Tolerancing (GD&T) principles to ensure design viability and dimensional accuracy for hardware enclosures.

CERTIFICATIONS

Safety Officer 2 (BOSH SO2) *April 2026*

Department of Labor and Employment (DOLE)

- Completed the 40-hour mandatory basic occupational safety training covering hazard identification, workplace risk control, and OSH standards compliance.

Certified Associate *[Month Year]*

Siemens Digital Industries Software

- Earned industry-standard credential verifying proficiency in Solid Edge 3D CAD modeling, assembly design, and digital drafting fundamentals.

Introduction to Cybersecurity *September 2026*

Cisco Networking Academy

- Achieved student-level credential covering cybersecurity fundamentals, common cyber threats and vulnerabilities, and personal and organizational protection practices.

SKILLS & INTERESTS

Programming & Software: C++, Java, Hardware Description Languages (HDL), Python, Object-Oriented Programming, Programming Logic & Design

Hardware & Engineering Tools: Siemens Solid Edge CAD, Arduino IDE, Microcontroller Programming, Circuit Analysis, Logic Circuit Design, Systems Administration

Networking & Security: Routing & Switching, Network Security Protocols, TCP/IP, Cybersecurity Fundamentals, Systems Administration, Digital Signal Processing

Safety & Compliance: Workplace Hazard Identification, Risk Assessment, OSH Standards (RA 11058 / DOLE DO 252-25)

Languages: English (Proficient), Filipino (Native)

Interests: Embedded Systems Prototyping, IoT Automation, Hardware Troubleshooting, Emerging Technology Trends